

Placement, Routing and the Border Contract

What decides where a tenant's data lives, how a request reaches the cell that is allowed to hold it, and what may cross a national border. **This is a directional document:** a first-pass position recorded so the platform can move, not a settled decision, stated as at 3 September 2026. Nothing on this sheet is built — the lab runs one server in one region — and no V3 decision of record exists yet. Messaging mechanics belong to LB-V3-L3-05; the placement record, allow-list text and country rules belong to L4.

PLACEMENT POLICY · WHAT DECIDES WHERE A TENANT'S DATA LIVES

- P

Tenant placement record
names the permitted cells
held by the control plane
proposed · no V3 ADR
- C

Country rule set
what a country demands
residency, retention, transit
open · legal to confirm
- D

Data class on each item
the class decides the border
set where the data is made
proposed · no V3 ADR
- A

One account, many cells
region is not a tenancy axis
one identity in every cell
proposed · no V3 ADR

policy is resolved before any traffic is placed

ROUTING · HOW A REQUEST REACHES THE CELL THAT MAY HOLD ITS DATA

- E

Country endpoint
one entry name per country
HTTPS, ends in the country
proposed · no V3 ADR
- R

Resolve before routing
tenant and policy read first
where you are decides nothing
proposed · no V3 ADR
- I

Cell ingress and web
one door, one asset origin
shell and plugins are local
proposed · no V3 ADR
- N

No silent re-routing
an outage never moves data
a regulated request waits
proposed · no V3 ADR

the request lands in one cell, carrying its data class

DATA CLASSES · THE CLASS DECIDES WHAT MAY CROSS A BORDER

- 1

Platform reference data
shared, owned by nobody
may be copied to each cell
proposed · no V3 ADR
- 2

Tenant business records
loads, trips, invoices
stays inside its own cell
proposed · no V3 ADR
- 3

Regulated country data
named by a country rule
no border path at all
open · legal to confirm
- 4

Operational telemetry
metrics, traces, masked logs
crosses only by policy
open · legal to confirm

only a named path may take a class across a border

BORDER CONTRACT · THE ONLY THREE WAYS ANYTHING CROSSES

- S

Signed policy and claims, in
the control plane writes to every cell
policy, account claims and catalogue entries only
proposed · no V3 ADR · carries no business payload
- G

Approved subjects, across
a gateway between two shared cells only
an allow-list names every subject that may cross
open · which subjects, and whether at all
- M

Named stream copy, across
a mirror or source, never on by default
chosen because a copy resumes after a break
open · which streams, and in which direction

- X

Sovereign border · nothing regulated crosses
no gateway, no mirror, no backup copy, no remote support session
only signed policy and account claims enter, and they carry no payload
open · legal confirmation required before this is compliance evidence

- !

Residency outranks convenience
policy decides placement; the cell boundary enforces it
where a person signs in decides nothing about where data lives
directional position as at 2026-09-03 · no V3 ADR of record

Read it downward. Policy chooses the cell, routing delivers the request to it, the data class decides whether anything may leave, and the border contract lists the only three ways out. **Built today:** nothing on this sheet — the lab is one server in one region. **Not shown:** subjects, accounts and grants, which are LB-V3-L3-05 ; the record schema, allow-list text and country rules, which are L4 .

- SCOPE

One question. Where may a tenant's data live, and what may cross a border? Availability and recovery are on sheet 2.
- DIRECTION

Directional. A first pass as at 2026-09-03 so work can start. Not a decision, and not compliance evidence.
- PLACEMENT

A **placement record** names the cells one tenant may use. It belongs to the tenant account, not to a person or a browser.
- CLASS

Every item carries a **data class**. The class, not the service, decides what happens at a border.
- ROUTING

The tenant is resolved **first**, then an ingress is chosen. Where a person happens to be decides nothing.
- LOCAL

The page, the app shell and the plugin registry are served from the **same cell** as the data they show.
- MIRROR

Durable history crosses only as a **named copy**. A copy is preferred over a live link because it resumes after a break.
- SOVEREIGN

A sovereign cell has **no** regulated path out, in any direction. Backups and support sessions are included in that.
- ACCOUNT

A tenant is **one account** in every permitted cell. The account carries the tenant; the deployment carries the region.
- OPEN

Every amber tile needs a named owner to confirm it. Ten open items are listed in the **L3-08 register**.

LEGEND

- Green boundary** — the policy that decides placement
- Solid tile** — built in the lab; there is none on this sheet
- Dashed grey tile** — proposed for V3, not built
- Red panel** — a path that is refused outright
- Red barred line** — a border nothing regulated crosses
- Blue boundary** — how a request reaches a cell
- Dashed grey tile** — proposed for V3, not built
- Green line** — a policy decision is applied
- Arrow head** — the direction of the relationship
- Cyan boundary** — the classes of data being placed
- Dashed amber tile** — open, a named owner must confirm
- Blue line** — a request travelling over HTTPS
- Line label** — names the relationship, never bare "uses"

- Amber boundary** — permitted cross-border paths
- Amber panel** — a rule that governs the whole sheet
- Amber dashed line** — data offered to a border rule

WORDS ON THIS SHEET

- cell** one region's complete runtime

region a place a cell is deployed in

tenant a marketplace-operating business

account the NATS tenant boundary

residency the rule saying where data may sit

data class a label deciding border behaviour

gateway a link joining two NATS clusters
- mirror / source** a copy of one stream elsewhere

sovereign kept inside one country by law

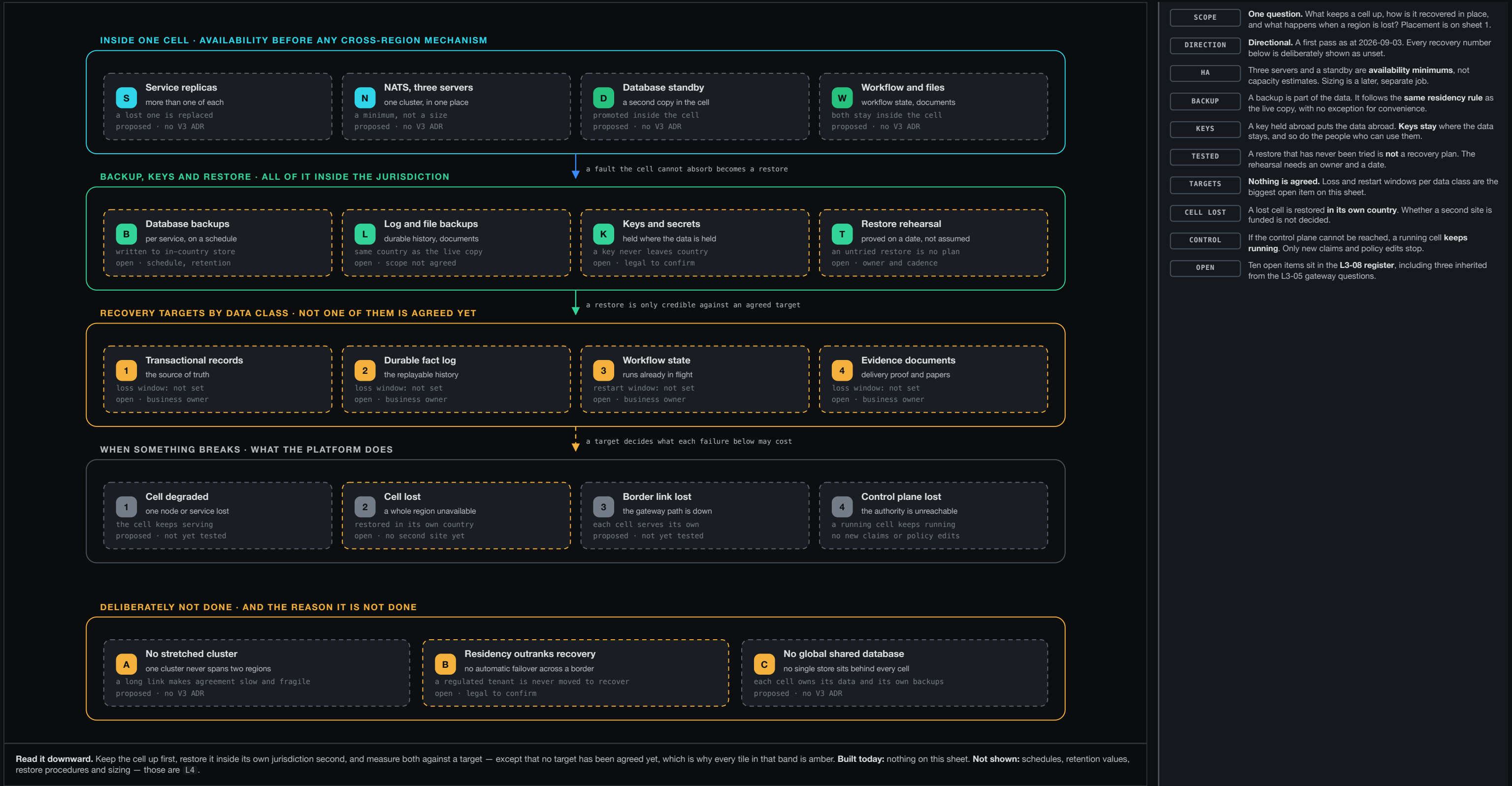
ingress the guarded door into a cell

HTTPS encrypted web traffic

ADR architecture decision record

Availability, Recovery and Failure by Jurisdiction

What keeps one cell running, how that cell is backed up and restored without leaving its jurisdiction, what each data class is allowed to lose, and what the platform does when a region, a border link or the control plane is lost. **This is a directional document:** a first-pass position stated as at 3 September 2026, not a settled decision. Nothing on this sheet is built, no recovery target has been agreed, and no V3 decision of record exists. Sizing, schedules and procedures belong to L4.



LEGEND

- | | | | | | | | |
|--|---|---|---|---|---|---|---|
| | Cyan boundary — availability inside one cell | | Green boundary — backup, keys and restore | | Amber boundary — unsettled, or deliberately refused | | Grey boundary — behaviour, not a component |
|  | Solid tile — built in the lab; there is none on this sheet |  | Dashed grey tile — proposed for V3, not built |  | Dashed amber tile — open, a named owner must confirm |  | Blue line — a fault escalating to the next answer |
|  | Green line — a restore measured against a target |  | Amber dashed line — a target governing a failure |  | Arrow head — the direction of the relationship | | Line label — names the relationship, never bare "uses" |

WORDS ON THIS SHEET

cell one region's complete runtime	rehearsal a restore tried on purpose, on a date
jurisdiction the country whose law applies	loss window how much recent data may be lost
HA high availability, running through a fault	restart window how long a return may take
standby a second database copy, ready to take over	gateway a link joining two NATS clusters
promote making the standby the live copy	control plane the platform's own authority services
restore rebuilding from a backup	ADR architecture decision record